Linear Regression Diagnostic

Model Diagnostics

How do we check whether our model meets its assumptions?

Residual: Difference between the observed value and the model's predicted value.

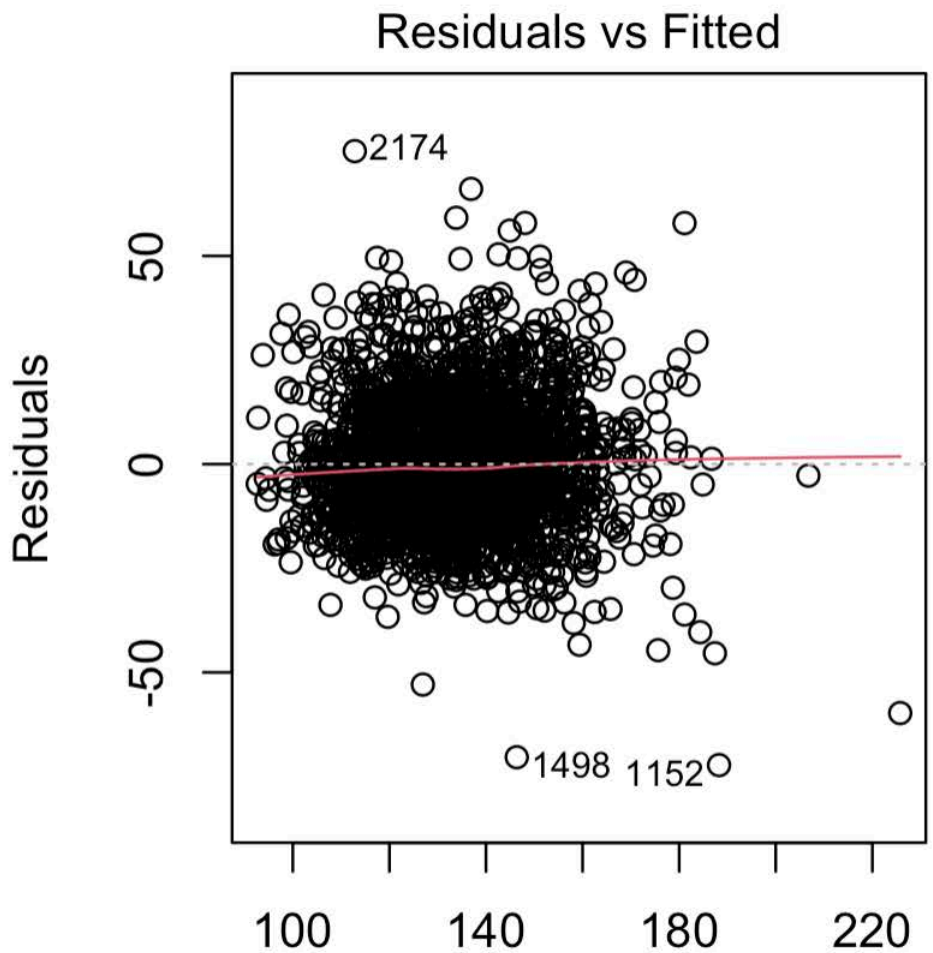
- Residuals should be approximately normally distributed, particularly for statistical inference in small samples.

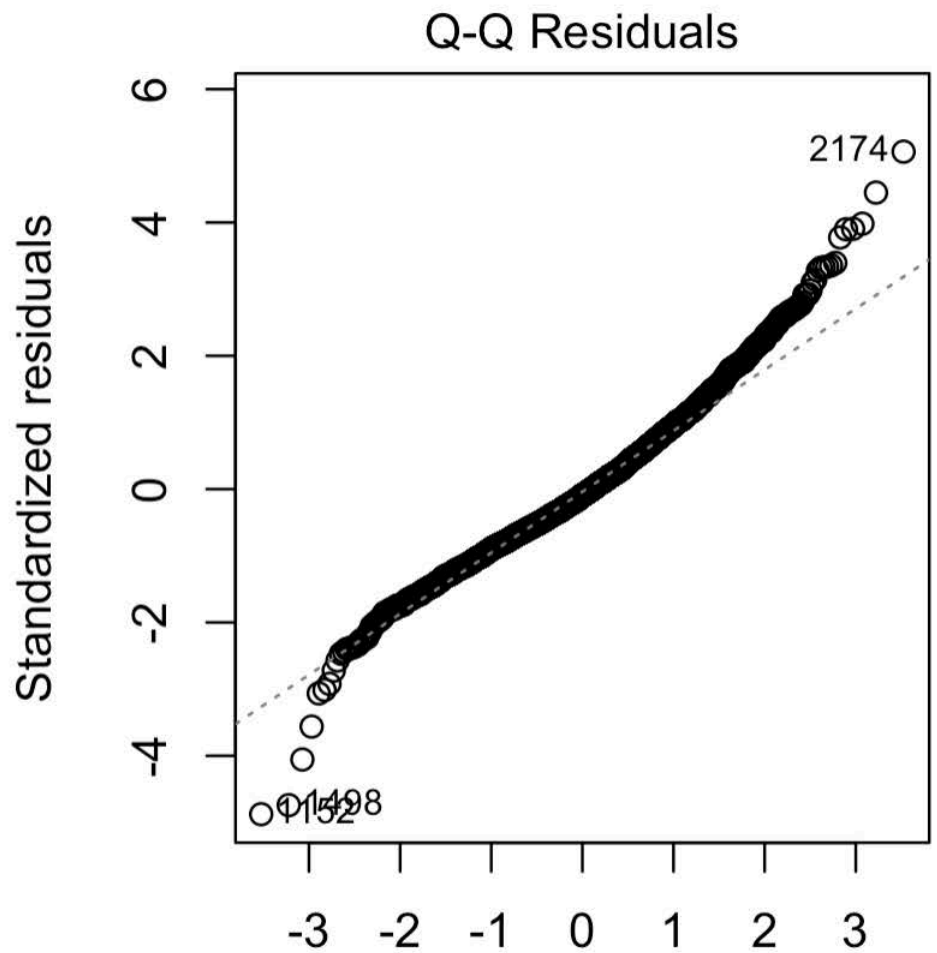
- Residuals should be scattered randomly around zero, with no clear patterns or changes in spread when plotted against predicted values.

Demo

Checking residuals

```
plot(model_multi, which = 1)   # Residuals vs. Fitted
plot(model_multi, which = 2)   # QQ plot
```







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